10/4/23

Mathematics

1. $p(x) = x^2 - 2x - (7P+3)$

$p(-4) = 16 - 2(-4) - (7P+3)$

$= 16 + 8 - 7P - 3$

$0 = 24 - 3 - 7P$

$= 21 - 7P$

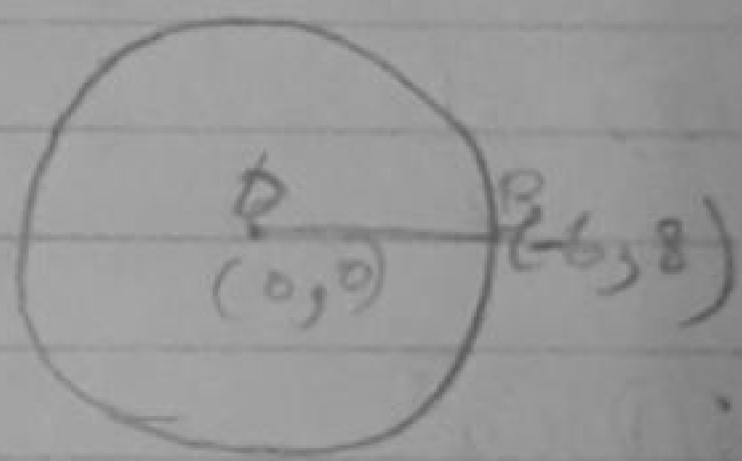
$7P = 21$

$P = 3$

(1) 3

2. (4) No solution.

3. Radius = OB



$OB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(6-0)^2 + (8-0)^2}$

$= \sqrt{6^2 + 8^2}$

$= \sqrt{36 + 64}$

$= \sqrt{100}$

$= 10$ units

(1) 10 units.

4. (2) Proportional

5. $\sqrt{3}\tan\theta = 2\sin\theta$

$\sqrt{3}\frac{\sin\theta}{\cos\theta} = 2\sin\theta$

$\frac{\sqrt{3}\cdot 1}{2\cos\theta} = \cancel{\theta}$

$\frac{\sqrt{3}}{2} = \cos\theta$

1

$\theta = 30^\circ$

$\sin^2\theta - \cos^2\theta$

$= \cancel{\left(\frac{1}{2}\right)^2} = \sin^2 30^\circ - \cos^2 30^\circ$

$= \left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2$

$= \frac{1}{4} - \frac{3}{4}$

$= \frac{-2}{4} = \frac{-1}{2}$

(2) $\frac{-1}{2}$

6(1) $x^3 y^3$

7. $4x^2 - 2x + (k-4)$

$\alpha + \beta = \frac{1}{2} \alpha\beta$

$\alpha + \beta = \frac{2}{4} \left(\frac{-b}{a}\right)$

$= \frac{1}{2}$

$\alpha\beta = \frac{c}{a}$

$= \frac{k-4}{4}$

$36 \quad 27$

$2)\,72 \quad 2)\,54$

$6 \quad 4$

$12 \quad 4$

108

$2^2 3^3 = 4 \times 27$

$2^3 3^2 = 8 \times 9$

$= 72$

2	108	72
2	54	36
3	27	18
3	9	6
3	3	2
2	1	2

$x^2 y^3$ 1, 1

$\alpha + \beta = \frac{1}{2} \alpha\beta$

$\frac{1}{2} = \frac{1}{2}\left(\frac{k-4}{4}\right)$

$\frac{1}{2} = \frac{k-4}{8}$

$\frac{8}{2} = k-4$

$k-4 = 4$
$k = 4+4$
$= 8$

8. $A(0,0)$; $B(3,\sqrt{3})$; $C(3,x)$

$AB = BC = CA$

$AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
$= \sqrt{(3-0)^2+(\sqrt{3}-0)^2}$
$= \sqrt{3^2+3}$
$= \sqrt{9+3}$
$= \sqrt{12}$

$BC = \sqrt{(3-3)^2+(x-\sqrt{3})^2}$
$\sqrt{12} = \sqrt{0+x^2-2\sqrt{3}x+3}$
$(\sqrt{12})^2 = (\sqrt{x^2-2\sqrt{3}x+3})^2$
$\cancel{12} = x^2-2\sqrt{3}x+3 = 12$
$x^2-2\sqrt{3}x-9=0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2\sqrt{3} \pm \sqrt{(2\sqrt{3})^2 - 4(1)(-9)}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{12 + 36}}{2}$$

$$= \frac{2\sqrt{3} \pm \sqrt{48}}{2}$$

$$= \frac{2\sqrt{3} \pm 4\sqrt{3}}{2}$$

$$= \frac{2\sqrt{3} - 4\sqrt{3}}{2} \quad ; \quad \frac{2\sqrt{3} + 4\sqrt{3}}{2}$$

$$= \frac{-\cancel{2}\sqrt{3}}{\cancel{2}} \quad ; = \frac{\overset{3}{\cancel{6}}\sqrt{3}}{\cancel{2}}$$

$$= -\sqrt{3} \quad ; = 3\sqrt{3}$$

$x = -\sqrt{3}$ is rejected.

$x = 3\sqrt{3}$

9. $2px + 3y = 7$
$2x + y = 6$

$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \Rightarrow$ one solution

11. $\tan\theta = \frac{1}{2}$

$\frac{\sin\theta}{\cos\theta} = \frac{1}{2}$

$\sin\theta = \frac{\cos\theta}{2}$

$\frac{1}{2}\sin\theta\ \cos\theta = \frac{1}{5}$

$\frac{1}{2} \times \frac{\cos\theta}{2} \times \cos\theta = \frac{1}{5}$

$\frac{\cos^2\theta}{4} = \frac{1}{5}$

$\cos^2\theta = \frac{4}{5}$

$\cos\theta = \frac{2}{\sqrt{5}}$

$\sin\theta = \frac{\cancel{2}}{\sqrt{5}} \times \frac{1}{\cancel{2}}$

$= \frac{1}{\sqrt{5}}$

12. $5^4 \times 2^3 = 5000$

$n = 3$

$m = 4$

$$\frac{257 \times 2}{5^4 \times 2^4}$$

$$= \frac{514}{10,000}$$

$= 0.0514$

13. assume = $\sqrt{3}$ is rational.

$\sqrt{3} = \frac{a}{b}$ (a and b are coprime)

$\cancel{\sqrt{3}b = a}$

$$3 = \frac{a^2}{b^2}$$

$3b^2 = a^2$

$a = 3c$ (1)

$\cancel{(3c)^2 = 3b^2}$

$\cancel{9c^2 = 3b^2}$

$\cancel{9c^2 - 3b^2 = 0}$

$\cancel{\frac{1}{3}b^2 = c^2}$

$(3c)^2 = 3b^2$

$3 \; 9c^2 = 3b^2$ (2)

b is divisible by 3

Since a and b are divisible by 3. They are not co-prime. Hence $\sqrt{3}$ is

(7)

an irrational number.

14. $p(x) = 2x^2 - 3x + 7$

$\alpha + \beta = \frac{3}{2}$

$\alpha \cdot \beta = \frac{7}{2}$

$\frac{1}{2\alpha - 3} + \frac{1}{2\beta - 3}$

$\frac{2\beta - 3 + 2\alpha - 3}{4\alpha\beta - 6\alpha - 6\beta + 9}$

$= \frac{2(\alpha + \beta) - 6}{-2(3\alpha + 3\beta - 4\alpha\beta) + 9}$

$= \frac{\cancel{2}\left(\frac{3}{\cancel{2}}\right) - 6}{-2\left(3(\alpha + \beta) - \cancel{4}^{2}\left(\frac{7}{\cancel{2}}\right)\right) + 9}$

$= \frac{3 - 6}{-2\left(3\left(\frac{3}{2}\right) - 14\right) + 9}$

$= \frac{-3}{\cancel{-2}\left(\frac{9 - 28}{\cancel{2}}\right) + 9}$

$= \frac{-3}{\cancel{-9} + 28 \cancel{+9}} = -3/28$

8

$3x - (a+1)y = -(2b-1)$
$5x + (1-2a)y = 3b$

$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow$ infinitely many solutions

$\frac{3}{5} = \frac{-a-1}{1-2a} = \frac{2b-1}{3b}$

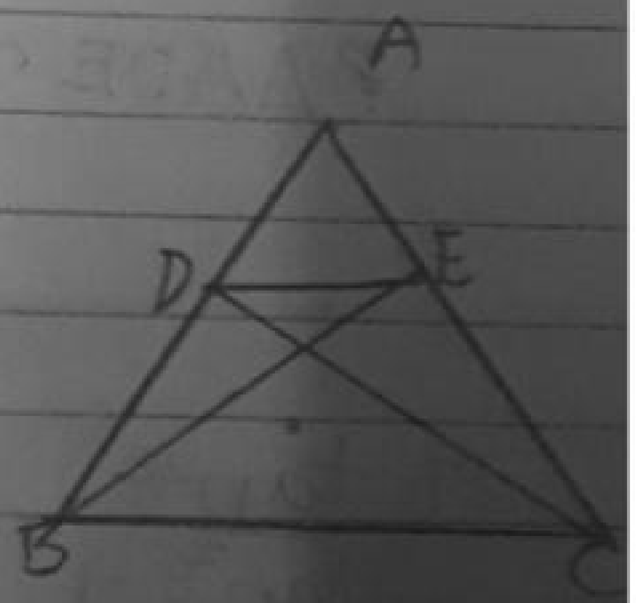
$\frac{-a-1}{1-2a} = \frac{3}{5}$

$-5(a+1) = 3(1-2a)$
$-5a - 5 = 3 - 6a$
$-5a + 6a = 3 + 5$
$a = 8$

$\frac{2b-1}{3b} = \frac{3}{5}$

$5(2b-1) = 3(3b)$
$10b - 5 = 9b$
$10b - 9b = 5$
$b = 5$

16 Given
$\Delta ABE \cong \Delta ACD$.



To Prove that
$\Delta ADE \sim \Delta ABC$

Statement	Reason
$AB = AC$	Given
$AB - DB = AC - AE$ $AD = AE$	If equals are subtracted from equals, the wholes are equal
$\frac{AD}{AB} = \frac{AE}{AC}$ (1)	Basic Proportionality Theorem
$\frac{DB}{AB} = \frac{EC}{AC}$ (2)	Basic Proportionality Theorem
$\frac{DB}{AB} = \frac{AD}{AB}$	
$\frac{AE}{AC} = \frac{EC}{AC}$	
$\frac{AD}{DB} = \frac{AE}{EC}$	Basic Proportionality Theorem
$DE \parallel BC$	
$\angle A = \angle A$	Common angle
$\triangle ADE \sim \triangle ABC$	By SAS similarity criteria

17. $$\frac{\tan^2\theta}{\tan^2\theta - 1} + \frac{\operatorname{cosec}^2\theta}{\sec^2\theta - \operatorname{cosec}^2\theta} = \frac{1}{\sin^2\theta - \cos^2\theta}$$

$$\frac{\sin^2\theta \times 1}{\cos^2\theta \times \frac{\sin^2\theta}{\cos^2\theta} - 1} + \frac{1 \times 1}{\sin^2\theta \times \frac{1}{\cos^2\theta} - \frac{1}{\sin^2\theta}}$$

$$\frac{\sin^2\theta}{\cos^2\theta \times \frac{\sin^2\theta - \cos^2\theta}{\cos^2\theta}} + \frac{1}{\sin^2\theta \times \frac{\sin^2\theta - \cos^2\theta}{\sin^2\theta\cos^2\theta}}$$

$a = \frac{c}{c}$

$a = \frac{xc}{b}$

$$\frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta - \cos^2\theta}$$

$$\frac{1}{\sin^2\theta - \cos^2\theta} = \frac{1}{\sin^2\theta - \cos^2\theta}$$

LHS = RHS

18. 3 : 2

A (-3, 7) C (1, 1) B (x, y)

$$B = \left(\frac{x_2 m_1 + x_1 m_2}{m_1 + m_2} ; \frac{y_2 m_1 + y_1 m_2}{m_1 + m_2}\right)$$

$$= \left(\frac{1 \times 3 + (-2)\,2}{3+2} ; \frac{1 \times 3 + 7 \times 2}{3+2}\right)$$

$$= \left(\frac{3-4}{5}, \frac{3+14}{5}\right)$$

$$= \left(\frac{-1}{5}, \frac{17}{5}\right)$$

19. $\frac{2x}{y+1} = 1$

$2x = y + 1$

$2x - y = 1$. (1)

$$\frac{x+4}{2y} = \frac{1}{2}$$

$$x+4 = \frac{\cancel{2}y}{\cancel{2}}$$

$$x - y = -4 \quad (2)$$

$$2x - y = 1$$

$$x - y = -4$$

$$x = 5$$

$$-y = -4-5$$

$$= -9$$

$$y = 9$$

$$\frac{x}{y} = \frac{5}{9}$$

21. $$\frac{\sin 30^\circ + 2\cos 45^\circ + \tan^2 60^\circ}{\frac{1}{2} \times \cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$$

$$= \frac{\frac{1}{2} + \cancel{2} \times \frac{1}{\cancel{\sqrt{2}}} + \sqrt{3}}{\frac{1}{2} \times 1 + \left(\frac{\sqrt{3}}{2}\right)^2 + 1^2}$$

$$= \frac{\frac{1+2\sqrt{2}+2\sqrt{3}}{2}}{\frac{1}{2} + \frac{3}{4} + 1} = \frac{\frac{1+2\sqrt{2}+2\sqrt{3}}{\cancel{2}}}{\frac{2+3+4}{\cancel{4}\,2}}$$

$$\frac{(1+2\sqrt{2}+2\sqrt{3})\times 2}{9}$$

$= \frac{\cancel{2(1+2\sqrt{2}+2\sqrt{3})}}{}$

$$\frac{2+4\sqrt{2}+2\sqrt{3}}{9}$$

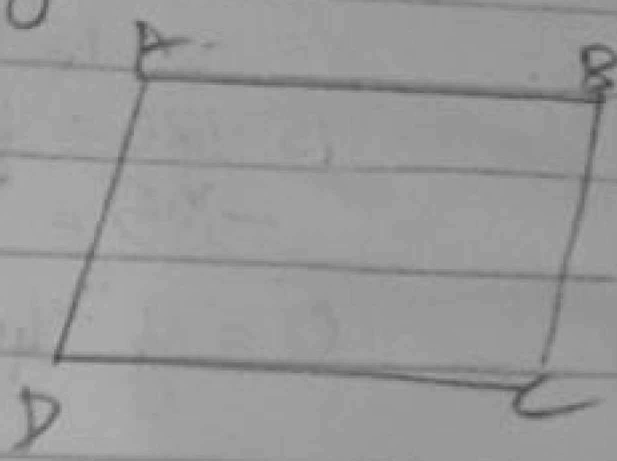
$a \div \frac{b}{c}$

$a \times \frac{c}{b}$

22. $A(1,2)$; $B(2,3)$; $C(6,5)$; $D(x,y)$

$AB = DC$

$AD = BC$



$AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(2-1)^2+(3-2)^2}$

$= \sqrt{1+1}$

$= \sqrt{2}$

$DC = \sqrt{(x-6)^2+(y-5)^2}$

$\sqrt{2} = \sqrt{x^2-12x+36+y^2-10y+25}$

$(\sqrt{2})^2 = \left(\sqrt{x^2+y^2-12x-10y+61}\right)^2$

$2 = x^2+y^2-12x-10y+61$

$-59 = x^2-12x+y^2-10y$. (1)

$BC = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$

$= \sqrt{(6-2)^2+(5-3)^2}$

$= \sqrt{4^2+2^2}$

$= \sqrt{16+4}$

$= \sqrt{20}$

$AD = \sqrt{(x-1)^2+(y-2)^2}$

$(\sqrt{20})^2 = \left(\sqrt{x^2-2x+1+y^2-4y+4}\right)^2$

$20 = x^2-2x+y^2-4y+5$

$15 = x^2-2x+y^2-4y$ (2)

(13)

$-59 = x^2 - 12x + y^2 - 10y$

$= x(x - 12) + y(y - 10)$

$-59 = (x+y)(x-12)$

$x = 12 \quad \rightarrow \cancel{x = -y} \qquad x = -y$

$15 = 12^2 - 2(12) + y^2 - 4y$

$= 144 - 24 + y(y-4)$

$= 120 + y^2 - 4y$

$15 - 120 = y^2 - 4y$

$-105 = y^2 - 4y$

$0 = y^2 - 4y + 105$

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{4^2 - 4(1)(105)}}{2}$$

$1, 2, 4, 101$
202

$$= \frac{4 \pm \sqrt{16 - 420}}{2}$$

$$= \frac{4 \pm \sqrt{404}}{2}$$

$$= \frac{4 \pm \sqrt{4 \times 101}}{2}$$

$$= \frac{4 \pm 4\sqrt{101}}{2}$$

$$= \frac{4 - 4\sqrt{101}}{2} ; \quad \frac{4 + 4\sqrt{101}}{2}$$

$= \frac{\cancel{4}^{2}(1-\sqrt{101})}{\cancel{2}}$; $\frac{\cancel{4}^{2}(1+\sqrt{101})}{\cancel{2}}$

$= 2(1-\sqrt{101})$; $= 2(1+\sqrt{101})$

Q. (12) $x = 12$
$y = -12$

23. Circumference $= 1980m$
$A = 330 m/min$
$B = 198 m/min$
$C = 220 m/min$

$D = 1980$

$S = \frac{D}{T}$

$T = \frac{D}{S}$

$T_1 = \frac{198\cancel{0}}{33\cancel{0}} = 6$ min.

$T_2 = \frac{\cancel{1980}}{\cancel{198}} = 10$ min

$T_3 = \frac{198\cancel{0}}{22\cancel{0}} = 9$ min

HCF
~~LCM~~ of 6, 10, 9

$33 \times 6 = 198$

$\cancel{33 \times 10}$

$\cancel{33 \times 5 = 165}$

$22 \overline{)198}$

$\cancel{22 \times 4 = 88}$

$\cancel{22 \times 6 = 132}$

$22 \times 8 = 176$

$22 \times 9 = 198$

$$\begin{array}{r|l} 2 & 6, 10, 9 \\ 3 & 3, 5, 9 \\ 5 & 1, 5, 3 \\ 3 & 1, 1, 3 \\ & 1, 1, 1 \end{array}$$

$2 \times 3^2 \times 5 = 10 \times 9 = 90$ mins

$$\begin{array}{r|l} 2 & 6 \\ 3 & 3 \\ & 1 \end{array} \quad \begin{array}{r|l} 2 & 10 \\ 5 & 5 \\ & 1 \end{array} \quad \begin{array}{r|l} 3 & 9 \\ 3 & 3 \\ & 1 \end{array}$$

$$\begin{array}{r|l} 2 & 6, 10, 9 \\ 3 & 3, 5, 9 \\ 5 & 1, 5, 3 \\ 3 & 1, 1, 3 \\ & 1, 1, 1 \end{array}$$

= 90 mins = They will meet after 90 mins at starting point.

24. $2x + y = 70$ y = Father
$2y + x = 95$ x = Son

$x + 2y = 95$ $2x + 4y = 190$
$(2x + y = 70) \times 2$ $2x + y = 70$
$3y = 120$
$y = 40$

95
70
95
80
15

$x = 15$
Father = 40, Son = 15

25 $A(1,-7)$ $\quad m_1 : m_2 \quad$ $C(x,0)$ $\quad B(6,4)$

$$AC = \left(\frac{x_1 m_2 + x_2 m_1}{m_1+m_2}, \frac{y_1 m_2 + y_2 m_1}{m_1+m_2}\right)$$

$$= \left(\frac{1 \times m_2 + x m_1}{m_1+m_2}, \frac{-7m_2 + 0}{m_1+m_2}\right)$$

$$= \left(\frac{m_2 + m_1 x}{m_1+m_2}, \frac{-7m_2}{m_1+m_2}\right)$$

$CB = ($

26 $\operatorname{cosec}\theta - \sin\theta = m$

$\sec\theta - \cos\theta = n.$

$$\frac{1}{\sin\theta} - \sin\theta = m$$

$$\frac{1-\sin^2\theta}{\sin\theta} = m$$

$$\frac{\cos^2\theta}{\sin\theta} = m$$

$$\frac{1}{\cos\theta} - \cos\theta = n$$

$$\frac{1 - \cos^2\theta}{\cos\theta} = n \qquad n = \frac{\sin^2\theta}{\cos\theta}$$

$(m^2n)^{\frac{2}{3}} + (mn^2)^{\frac{2}{3}}$

$\left(\frac{\cos^{3}\theta}{\sin^{2}\theta} \times \frac{\sin^{2}\theta}{\cos\theta}\right)^{\frac{2}{3}} + \left(\frac{\cos^{2}\theta}{\sin\theta} \times \frac{\sin^{4}\theta}{\cos^{2}\theta}\right)^{\frac{2}{3}}$

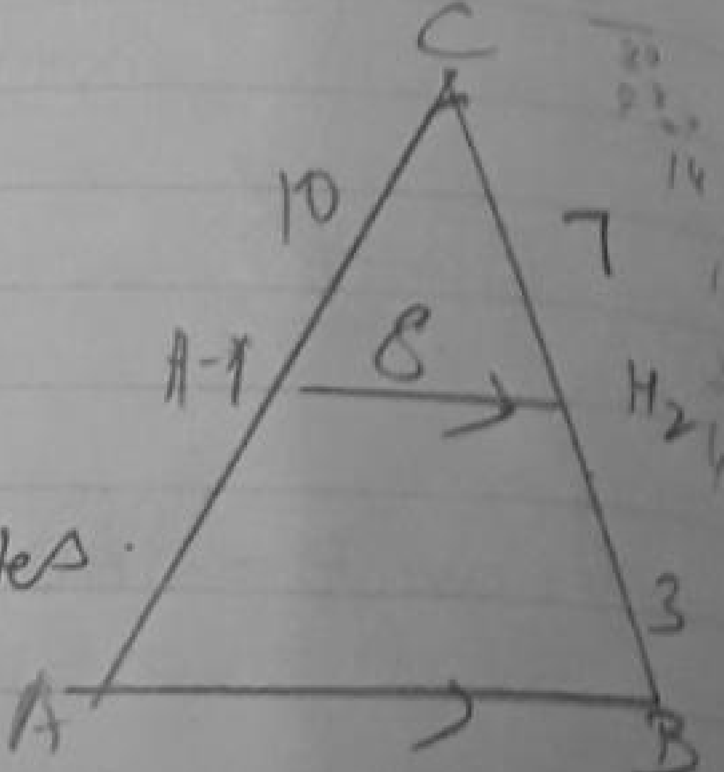
$\cos^{3\times\frac{2}{3}}\theta + \sin^{3\times\frac{2}{3}}$

$\cos^2\theta + \sin^2\theta$

$1 = 1$

LHS = RHS

Hence Proved.



27 (ii) 4.26 miles.

(iii) 18.2.

(iv) Hotel B to mountain = 5 miles.

$3x + 7x = 5$

$10x = 5$

$x = \frac{5}{10}$

$= \frac{1}{2}$

$\frac{7}{2} = 3.5$ miles.

2.6×7

18.2

28 (i) $P(2,5)$, $Q(x,y)$, $R(8,3)$

$m_1 : m_2$

$1 : 2$

$PQ = \left(\frac{x_1 m_2 + x_2 m_1}{m_1 + m_2}, \frac{y_1 m_2 + y_2 m_1}{m_1 + m_2}\right)$

$$= \left(\frac{2\times2 + 5\times1}{1+2}, \frac{5\times2+y}{1+2} \right)$$

$$= \left(\frac{4+5}{3}, \frac{10+y}{3} \right)$$

$$= \left(\frac{9}{3}, \frac{10+y}{3} \right)$$

$$= \left(3, \frac{10+y}{3} \right)$$

$x = 3$

$$QR = \left(\frac{x_1 m_2 + x_2 m_1}{m_1+m_2}, \frac{y_2 m_1 + y_1 m_2}{m_1+m_2} \right)$$

$$= \left(\frac{3\times2 + 8\times1}{1+2}, \frac{3\times1 + y\times2}{1+2} \right)$$

$$= \left(\frac{6+8}{3}, \frac{3+2y}{3} \right)$$

$=$

$Q = (4, 4)$

(ii) $PR = \sqrt{(8-2)^2 + (3-5)^2}$

$= \sqrt{6^2 + 2^2}$

$= \sqrt{36+4}$

$= \sqrt{40}$

(iii) $(3, 4)$

$$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$= \left(\frac{10}{2}, \frac{8}{2}\right)$$

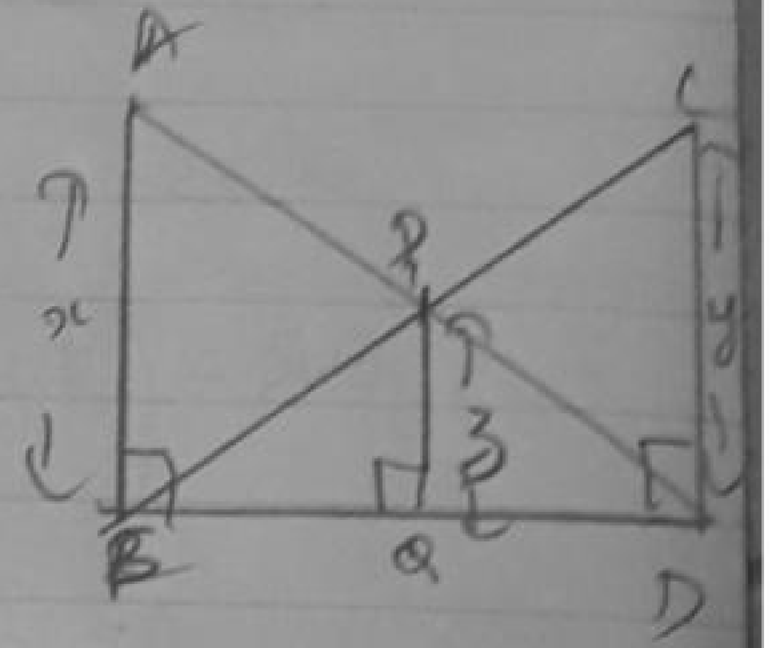
$$= (5, 4)$$

~~29. A − B = 100 km~~

30. Given − $\angle ABD = \angle CDB = \angle PQB = 90^\circ$

To Prove that

$$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$



Statement		Reason
$\angle ABD = \angle CDB = 90^\circ$	(R)	Given
$CD = CD$	(S)	Common side.
$AD = CB$	(H)	
$\triangle ABD \cong CBD$		RHS $\cong$
$x = y$		
$\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$		